

COMP 3311

DATABASE MANAGEMENT

SYSTEMS

LECTURE 13 EXERCISES

MONGODB DBMS:

SQL FIND METHOD

BOATRESERVATIONS MONGODB DATABASE

sailors collection

```
{sailorId: 10, sName: 'Emily', hkid: 'P936219(3)', age: 16},
{sailorId: 15, sName: 'Zoey', hkid: 'K091876(5)', rating: null, age: 17},
{sailorId: 22, sName: 'Dustin', hkid: 'A298456(6)', rating: 7, age: 45,
  reservations: [{boatId: 103, rDate: '08-SEP-25'},
                 {boatId: 104, rDate: '07-OCT-25'},
                 {boatId: 101, rDate: '10-OCT-25'},
                 {boatId: 102, rDate: '10-NOV-25'}]},
{sailorId: 29, sName: 'Brutus', hkid: 'D746239(0)', rating: 1, age: 33},
{sailorId: 31, sName: 'Lucy', hkid: 'P930281(5)', rating: 10, age: 55,
  reservations: [{boatId: 102, rDate: '10-OCT-25'},
                 {boatId: 103, rDate: '06-NOV-25'},
                 {boatId: 102, rDate: '12-NOV-25'}]},
{sailorId: 32, sName: 'Andy', hkid: 'M028391(4)', rating: 8, age: 25},
{sailorId: 58, sName: 'Rachael', hkid: 'A819204(3)', rating: 10, age: 17},
{sailorId: 64, sName: 'Horatio', hkid: 'F710943(2)', rating: 7, age: 35,
  reservations: [{boatId: 101, rDate: '05-SEP-25'},
                 {boatId: 102, rDate: '08-SEP-25'}]},
{sailorId: 71, sName: 'Zorba', hkid: 'P910562(4)', rating: 1, age: 16},
{sailorId: 74, sName: 'Horatio', hkid: 'T810463(1)', rating: 9, age: 35,
  reservations: [{boatId: 103, rDate: '08-SEP-25'}]},
{sailorId: 85, sName: 'Amy', hkid: 'B719037(8)', rating: 3, age: 25},
{sailorId: 95, sName: 'Bob', hkid: 'E012735(5)', rating: 3, age: 63},
{sailorId: 99, sName: 'Chris', hkid: 'H814243(7)', rating: 10, age: 30,
  reservations: [{boatId: 102, rDate: '08-AUG-25'}]}
```

boats collection

```
{boatId: 101, bName: 'Interlake', color: 'blue', bType: 'dinghy',
  reservations: [{sailorId: 64, rDate: '05-SEP-25'},
                 {sailorId: 22, rDate: '10-OCT-25'}]},
{boatId: 102, bName: 'Interlake', color: 'red', bType: 'sloop',
  reservations: [{sailorId: 64, rDate: '08-SEP-25'},
                 {sailorId: 31, rDate: '10-OCT-25'},
                 {sailorId: 22, rDate: '10-NOV-25'},
                 {sailorId: 31, rDate: '12-NOV-25'},
                 {sailorId: 99, rDate: '08-AUG-25'}]},
{boatId: 103, bName: 'Clipper', color: 'green', bType: 'dinghy',
  reservations: [{sailorId: 22, rDate: '08-SEP-25'},
                 {sailorId: 74, rDate: '08-SEP-25'},
                 {sailorId: 31, rDate: '06-NOV-25'}]},
{boatId: 104, bName: 'Marine', color: 'red', bType: 'catamaran',
  reservations: [{sailorId: 22, rDate: '07-OCT-25'}]},
{boatId: 105, bName: 'Serenity', color: 'cyan'}
```



EXERCISE 1

Find sailors who have a rating of 8 or greater.
Include in the query result objects fields sailor name, age and rating.

Query {sName: 'Lucy', age: 55, rating: 10} {sName: 'Andy', age: 25, rating: 8}
result: {sName: 'Rachael', age: 17, rating: 10} {sName: 'Horatio', age: 35, rating: 9}
{sName: 'Chris', age: 30, rating: 10}

```
db.sailors.find({rating: {$gte: 8}},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

EXERCISE 2

Find sailors who have a rating of 8 or greater and are 30 years old or younger.
Include in the query result objects fields sailor name, age and rating.

Query {sName: 'Andy', age: 2, rating: 85} {sName: 'Chris', age: 30, rating: 10}
result: {sName: 'Rachael', age: 17, rating: 10}

```
db.sailors.find({rating: {$gte: 8}, age: {$lte: 30}},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

```
db.sailors.find({$and: [{rating: {$gte: 8}}, {age: {$lte: 30}}]},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

EXERCISE 3

Find sailors between 20 and 30 years old who have not reserved any boat. Include in the query result objects fields sailor name, age and rating.

Query result: {sName: 'Andy', age: 25, rating: 8} {sName: 'Amy', age: 25, rating: 3}

Is this a correct solution?

No! Why?

```
db.sailors.find({age: {$gte: 20}, age: {$lte: 30}, reservations: {$exists: false}},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

Query {sName: 'Emily', age: 16} {sName: 'Zoey', age: 17, rating: null}
result: {sName: 'Andy', age: 25, rating: 8} {sName: 'Zorba', age: 16, rating: 1}
 {sName: 'Amy', age: 25, rating: 3} {sName: 'Rachael', age: 17, rating: 10}

The implicit **and** condition has two predicates on the same field age.
Only the last predicate is applied!

```
db.sailors.find({$and: [{age: {$gte: 20}}, {age: {$lte: 30}}, {reservations: {$exists: false}}]},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

```
db.sailors.find({age: {$gte: 20, $lte: 30}, reservations: {$exists: false}},  
  {_id: 0, sName: 1, age: 1, rating: 1});
```

Use either **\$and** operator or a list of conditions
when conjunctive predicates are on the same field.

EXERCISE 4

Find all sailors.
Include in the query result objects fields sailor name, rating and age replacing null and absent ratings with 'none'.

Query {sName: 'Emily', age: 16, rating: 'none'} {sName: 'Zoey', age: 17, rating: 'none'}
result: {sName: 'Dustin', age: 45, rating: 7} {sName: 'Brutus', age: 33, rating: 1}
 {sName: 'Lucy', age: 55, rating: 10} {sName: 'Andy', age: 25, rating: 8}
 {sName: 'Rachael', age: 35, rating: 10} {sName: 'Horatio', age: 35, rating: 7}
 {sName: 'Zorba', age: 16, rating: 1} {sName: 'Horatio', age: 35, rating: 9}
 {sName: 'Amy', age: 25, rating: 3} {sName: 'Bob', age: 63, rating: 3}
 {sName: 'Chris', age: 30, rating: 10}

Is this a
correct
solution?

No! Why?

~~db.sailors.find({
 { _id: 0, sName: 1, age: 1, rating: { \$cond: { if: { \$ne: ['\$rating', null]},
 then: '\$rating',
 else: 'none' }}});~~

Query {sName: 'Emily', age: 16} {sName: 'Zoey', age: 17, rating: 'none'}
result: {sName: 'Dustin', age: 45, rating: 7} {sName: 'Brutus', age: 33, rating: 1}
 :
 :

 Will not replace absent fields.

EXERCISE 4 (CONT'D)

Find all sailors.

Include in the query result objects fields sailor name, rating and age replacing null and absent ratings with 'none'.

Is this a
correct
solution?

No! Why?

```
db.sailors.find({  
  {_id: 0, sName: 1, age: 1, rating: { $cond: { if: { $exists: ['$rating', true] },  
    then: '$rating',  
    else: 'none' }}}},
```

Query error: Unrecognized expression '\$exists'

 **Cannot use \$exists within a \$cond operator in the find method.**

EXERCISE 4 (CONT'D)

Find all sailors.

Include in the query result objects fields sailor name, rating and age replacing null and absent ratings with 'none'.

```
db.sailors.find({ },  
  {_id: 0, sName: 1, age: 1, rating: {$ifNull: ['$rating', 'none']}});
```

```
db.sailors.find({ },  
  {_id: 0, sName: 1, age: 1, rating: {$cond: {if: {$gt: ['$rating', 0]},  
    then: '$rating',  
    else: 'none'}}});
```

EXERCISE 5

Find sailors who have reserved a boat but have not reserved boat 104.
Include in the query result objects fields sailor id and name.

Query {sailorId: 31, sName: 'Lucy'} {sailorId: 64, sName: 'Horatio'}
result: {sailorId: 74, sName: 'Horatio'} {sailorId: 99, sName: 'Chris'}

```
db.sailors.find({'reservations.boatId': {$ne: 104}, reservations: {$exists: true}},  
  {_id: 0, sailorId: 1, sName: 1});
```

```
db.sailors.find({$and: [{'reservations.boatId': {$ne: 104}}, {reservations: {$ne: null}}]},  
  {_id: 0, sailorId: 1, sName: 1});
```

 **Quotes are required around field names
when using the dot notation.**

EXERCISE 6

Find sailors who have reserved both boat 101 and boat 103.
Include in the query result objects fields sailor id, name and rating.

Query result: {sailorId: 22, sName: 'Dustin', rating: 7}

Is this a
correct
solution?

No! Why?

~~db.sailors.find({'reservations.boatId': 101, 'reservations.boatId': 103},
{_id: 0, sailorId: 1, sName: 1, rating: 1});~~

Query {sailorId: 22, sName: 'Dustin', rating: 7}
result: {sailorId: 31, sName: 'Lucy', rating: 10}
{sailorId: 74, sName: 'Horatio', rating: 9}

☞ **Sailors who have reserved boat 103.**

The implicit and condition is on the **same field** reservations.boatId.

☞ **Only the last predicate is applied!**

~~db.sailors.find({'reservations.boatId': {\$eq: 101, \$eq: 103}},
{_id: 0, sailorId: 1, sName: 1, rating: 1});~~

The same result as above since an implicit and condition, *when applied to array fields*, is evaluated differently than when applied to non-array fields.

(See next lecture.)

EXERCISE 6 (CONT'D)

Find sailors who have reserved both boat 101 and boat 103.
Include in the query result objects fields sailor id, name and rating.

```
db.sailors.find({$and: [{'reservations.boatId': 101}, {'reservations.boatId': 103}]},  
  {_id: 0, sailorId: 1, sName: 1, rating: 1});
```

Need to use the `$and` operator when predicates are on the same array field.

EXERCISE 7

Find boats that have been reserved.
Include in the query result objects fields boat id, name and an array field named "reservations" whose elements are objects, one for each reservation of the boat, with field sailor id.

Query {boatId: 101, bName: 'Interlake', reservations: [{sailorId: 22}, {sailorId: 64}]}

result: {boatId: 102, bName: 'Interlake', reservations: [{sailorId: 22}, {sailorId: 31}, {sailorId: 31}, {sailorId: 64}, {sailorId: 99}]}

{boatId: 103, bName: 'Clipper', reservations: [{sailorId: 22}, {sailorId: 31}, {sailorId: 74}]}

{boatId: 104, bName: 'Marine', reservations: [{sailorId: 22}]}

```
db.boats.find({reservations: {$ne: null}},  
  {_id: 0, boatId: 1, bName: 1, 'reservations.sailorId': 1});
```

```
db.boats.find({reservations: {$exists: true}},  
  {_id: 0, boatId: 1, bName: 1, 'reservations.sailorId': 1});
```